Lecture 1 · Vectors

Part 1 of 2 · All lecture topics and distinct examples; repeated reveal slides combined.

A vector is an **ordered list of real numbers**, $\mathbf{v} \in \mathbb{R}^n$; geometrically, it is a displacement with length and direction. Knitting connection: $(3,1,3,1)^T$ can record pink-stitch counts in four six-stitch sections. These counts omit stitch order and tension; geometry compares records.

1. Vector representation, addition, and scaling

Lecture examples (pp. 4–6): $\mathbf{v} = (5,2)^T \in \mathbb{R}^2$, $\mathbf{w} = (1,7,3)^T \in \mathbb{R}^3$. A house becomes $(3,2,2,2008,150000)^T$: bedrooms, bathrooms, floors, year, price. Coordinate meanings and units matter. Adding a coordinate, as in $(1,2)^T \to (1,2,3)^T$, changes the coordinate space.

With standard directions $\mathbf{e}_1 = (1,0)^T$, $\mathbf{e}_2 = (0,1)^T$, $(a,b)^T = a\mathbf{e}_1 + b\mathbf{e}_2$. In $\mathbb{R}^3$, use $\mathbf{e}_1 = (1,0,0)^T$, $\mathbf{e}_2 = (0,1,0)^T$, $\mathbf{e}_3 = (0,0,1)^T$ to write $(a,b,c)^T = a\mathbf{e}_1 + b\mathbf{e}_2 + c\mathbf{e}_3$.

Addition: $(\mathbf{u}+\mathbf{v})_i = u_i + v_i$ for matching dimensions. Draw it head-to-tail. **Scalar multiplication:** $(\lambda\mathbf{v})_i = \lambda v_i$; positive scalars stretch/shrink, negative scalars reverse direction, zero gives zero.

Lecture examples · Addition and scalar multiplication (pp. 8–9)

For $\mathbf{x} = (0,2)^T$, $\mathbf{y} = (2,2)^T$: $\mathbf{x}+\mathbf{y} = (2,4)^T$ and $\mathbf{x}-\mathbf{y} = (-2,0)^T$.
Separately, for $\mathbf{x} = (2,2)^T$: $2\mathbf{x} = (4,4)^T$, $\frac{1}{2}\mathbf{x} = (1,1)^T$, $-\mathbf{x} = (-2,-2)^T$.

2. Linear combinations: what can the vectors produce?

A **linear combination** is $c\mathbf{u} + d\mathbf{v} + e\mathbf{w}$, with real coefficients. One nonzero direction produces a line through zero; two nonparallel directions produce a plane; three independent directions in $\mathbb{R}^3$ produce all of it. Redundant directions add no new reachable points; all-zero generators give only zero.

Lecture's geometric example (pp. 12–18): $\mathbf{u} = (1,2,4)^T$, $\mathbf{v} = (2,1,4)^T$, $\mathbf{w} = (9,8,7)^T$. The records $c\mathbf{u}$, $c\mathbf{u} + d\mathbf{v}$, and $c\mathbf{u} + d\mathbf{v} + e\mathbf{w}$ respectively fill a line, plane, and $\mathbb{R}^3$. Here w is outside the plane generated by u,v.

Lecture example · Plane membership (p. 19)

$$c(1,1,0)^T + d(0,1,1)^T = (c, c+d, d)^T.$$

Thus the plane is $y = x + z$: choose $c = x$, $d = z$. The source's $(1,2,3)^T$ lies outside because $2 \neq 1 + 3$. Compare coordinates to test membership; do not guess endlessly.

3. Dot product, angle, length, magnitude, and norm

Essential formulas · Real Euclidean vectors

Concept	Formula and condition
Dot/inner product	$\mathbf{u}\cdot\mathbf{v} = \langle \mathbf{u},\mathbf{v}\rangle = \mathbf{u}^T\mathbf{v} = \sum_{i=1}^n u_i v_i$; equal dimensions; **scalar** output
Length = magnitude = norm	$\|\mathbf{v}\| = \sqrt{\mathbf{v}^T\mathbf{v}} = \sqrt{\sum_i v_i^2}$
Unit vector	$\widehat{\mathbf{v}} = \mu\mathbf{v}$, $\mu = 1/\|\mathbf{v}\|$; then $\|\widehat{\mathbf{v}}\| = 1$, for $\mathbf{v} \neq \mathbf{0}$
Cosine formula	$\mathbf{u}^T\mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\|\cos\theta$
Angle between two vectors	$\theta = \arccos\left(\dfrac{\mathbf{u}^T\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}\right)$; both vectors nonzero

Dot-product properties (p. 23): commutative, $\mathbf{u}^T\mathbf{v} = \mathbf{v}^T\mathbf{u}$; distributive, $\mathbf{u}^T(\mathbf{v}+\mathbf{w}) = \mathbf{u}^T\mathbf{v} + \mathbf{u}^T\mathbf{w}$. The lecture's **not-associative** warning means an inner product returns a scalar: $\mathbf{u}\cdot(\mathbf{v}\cdot\mathbf{w})$ is not another vector dot product. Even interpreting the inner results as scalar multipliers generally gives $\mathbf{u}(\mathbf{v}^T\mathbf{w}) \neq (\mathbf{u}^T\mathbf{v})\mathbf{w}$.

Why orthogonality and cosine work (pp. 24–26): for perpendicular vectors, Pythagoras gives $\|\mathbf{u}-\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$. In general, compare the algebraic expansion with the cosine law:

$$\|\mathbf{u}-\mathbf{v}\|^2 = (\mathbf{u}-\mathbf{v})^T(\mathbf{u}-\mathbf{v}) = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\mathbf{u}^T\mathbf{v},$$

$$\|\mathbf{u}-\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\|\mathbf{u}\|\|\mathbf{v}\|\cos\theta.$$

Equating yields the cosine formula. **Orthogonal means dot product zero**; nonzero orthogonal vectors have angle 90°. Positive dot product means angle below 90°; negative means above it.

Lecture examples · Dot product, norm, and unit vector (pp. 22, 28–34)

For $\mathbf{u} = (3,4)^T$, $\mathbf{v} = (4,3)^T$: $\mathbf{u}^T\mathbf{v} = 3\cdot 4 + 4\cdot 3 = 24$. $\|\mathbf{u}\| = \sqrt{3^2+4^2} = 5$; likewise $\|\mathbf{v}\| = 5$. Thus $\mu = 1/5$, $\widehat{\mathbf{u}} = (3/5, 4/5)^T$. Applying the lecture's angle formula gives $\cos\theta = 24/25$, $\theta \approx 16.26^\circ$.

Exam warnings

Keep the square root in a norm. Zero has no normalized direction or defined angle. A raw dot product is not a cosine. In general $\|c\mathbf{v}\| = |c|\|\mathbf{v}\|$.

Continue to matrices and all matrix examples. *Source: Erisa Terolli, CS 556, Week 1 "Matrices and Vectors," pp. 1–34. Knitting connection and the numerical angle evaluation are added exposition.*

Lecture 1 · Matrices

Part 2 of 2 · Matrix definitions, every lecture example, and multiplication properties.

4. Matrices and special structures

$A = (a_{ij}) \in \mathbb{R}^{m \times n}$ means **m rows, n columns**; a_{ij} is row i, column j. A sampler table can use worked rows and marked sections as its two index roles. Equal numerical shapes alone do not make their physical meanings interchangeable.

Structure	Definition
Diagonal	Square; $a_{ij} = 0$ whenever $i \neq j$
Identity I_n	Diagonal ones, off-diagonal zeros; $I_n\mathbf{x} = \mathbf{x}$
Symmetric	Square; $A^T = A$, or $a_{ij} = a_{ji}$
Skew-symmetric	Square; $K^T = -K$, or $k_{ij} = -k_{ji}$; hence diagonal entries are zero

Lecture examples (pp. 37–39):

$$D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 8 \end{bmatrix}, \quad I_n = (\delta_{ij}), \quad S = \begin{bmatrix} 3 & 2 & 4 \\ 2 & 5 & 6 \\ 4 & 6 & 8 \end{bmatrix}.$$

Here $\delta_{ij} = 1$ if $i = j$, otherwise 0. Thus I is the lecture's general identity matrix; S is symmetric. A zero diagonal alone does **not** imply skew-symmetry.

5. Matrix addition, scalar multiplication, transpose, and trace

Essential formulas · Match the operation to the shape

Operation	Formula and shape
Matrix addition	$(A + B)_{ij} = a_{ij} + b_{ij}$; A and B must have the same shape
Scalar multiplication	$(\lambda A)_{ij} = \lambda a_{ij}$; same shape as A
Transpose	$(A^T)_{ji} = a_{ij}$; $m \times n \to n \times m$
Matrix trace	$\operatorname{tr}(A) = \sum_{i=1}^{n} a_{ii}$; square A

These entrywise formulas retain the lecture's general addition/scaling examples (pp. 40–41); those slides do not supply separate numerical matrices to calculate.

Lecture transpose example (p. 42): $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ gives $A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$. **Lecture trace example (p. 43):** for S above, $\operatorname{tr}(S) = 3 + 5 + 8 = 16$. Transpose swaps index roles; trace adds only the main diagonal.

6. Matrix-vector and matrix-matrix multiplication

Essential rule · Inner dimensions match; outer dimensions remain

For $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{n\times p}$:

$$(AB)_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj}, \qquad AB \in \mathbb{R}^{m\times p}.$$

Each entry is **row i of A dotted with column j of B**. For a vector, $A\mathbf{x} = \sum_{j=1}^{n} x_j \mathbf{a}_j$, where $\mathbf{a}_j$ are A's columns; $(A\mathbf{x})_i = \sum_j a_{ij}x_j$.

Lecture matrix-vector example (p. 45): with columns $\mathbf{u} = (1,-1,0)^T$, $\mathbf{v} = (0,1,-1)^T$, $\mathbf{w} = (0,0,1)^T$,

$$A\mathbf{x} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1\mathbf{u} + x_2\mathbf{v} + x_3\mathbf{w} = \begin{bmatrix} x_1 \\ x_2 - x_1 \\ x_3 - x_2 \end{bmatrix}.$$

The source asks which coefficients produce a given b: $A\mathbf{x} = \mathbf{b}$. That is the linear-system question developed in Lecture 2.

Lecture matrix product example (p. 46)

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 2 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad AB = \begin{bmatrix} 2 & 7 \\ 2 & 9 \end{bmatrix}.$$

Shapes: $(2\times 3)(3\times 2) = 2\times 2$. For instance, $(AB)_{12} = 1\cdot 2 + 2\cdot 1 + 3\cdot 1 = 7$.

Both interpretations (p. 47):

$$A[\mathbf{b}_1 \ \cdots \ \mathbf{b}_p] = [A\mathbf{b}_1 \ \cdots \ A\mathbf{b}_p],$$

$$\operatorname{row}_i(AB) = \operatorname{row}_i(A)B = \sum_{k=1}^{n} a_{ik} \operatorname{row}_k(B).$$

Thus product columns combine **columns of A**; product rows combine **rows of B**.

Crochet connection: plain single crochet consumes/produces $(1,1)^T$; an increase works two single crochets into one old stitch, giving $(1,2)^T$. Then $\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}(18,6)^T = (24,30)^T$: 24 consumed, 30 produced. Counts do not specify placement.

All lecture multiplication properties (p. 48)

For compatible shapes:

- **Associativity:** $(AB)C = A(BC)$, with A $m\times n$, B $n\times p$, C $p\times q$.
- **Left distributivity:** $(A+B)C = AC + BC$.
- **Right distributivity:** $A(C+D) = AC + AD$.
- **Identity:** $I_m A = A = AI_n$ for A $m\times n$.

In the distributive identities, A and B are $m\times n$, C and D are $n\times p$.

Exam warnings

Matrix multiplication generally **does not commute**: $AB \neq BA$. $AB\mathbf{x}$ applies B first. Transpose is not generally inverse; $(AB)^T = B^T A^T$ reverses order. A trace is not the sum of all matrix entries.

Back to vectors. *Source: Erisa Terolli, CS 556, Week 1 "Matrices and Vectors," pp. 35–48; original slides. Crochet connection and additional exam warnings are added exposition.*